

Task 1: Iris Flower Classification

Objective

Build a Decision Tree classifier to predict Iris flower species.

Steps

1. Load dataset
2. Explore data
3. Preprocess
4. Train model
5. Evaluate performance

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report

df = pd.read_csv("Iris.csv")
display(df.head())
display(df.describe())
print(df.info())
print(df.isnull().sum())

if "Id" in df.columns:
    df = df.drop("Id", axis=1)

X = df.drop("Species", axis=1)
y = df["Species"]

X_train, X_test, y_train, y_test =
train_test_split(X, y, test_size=0.2, random_state=42)

model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)

pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, pred))
print(confusion_matrix(y_test, pred))
print(classification_report(y_test, pred))

plt.figure(figsize=(6, 4))
plt.bar(X.columns, model.feature_importances_)
```

```
plt.title("Feature Importance")  
plt.show()
```

Conclusion

The Decision Tree classifier achieved high accuracy and successfully classified Iris flowers.